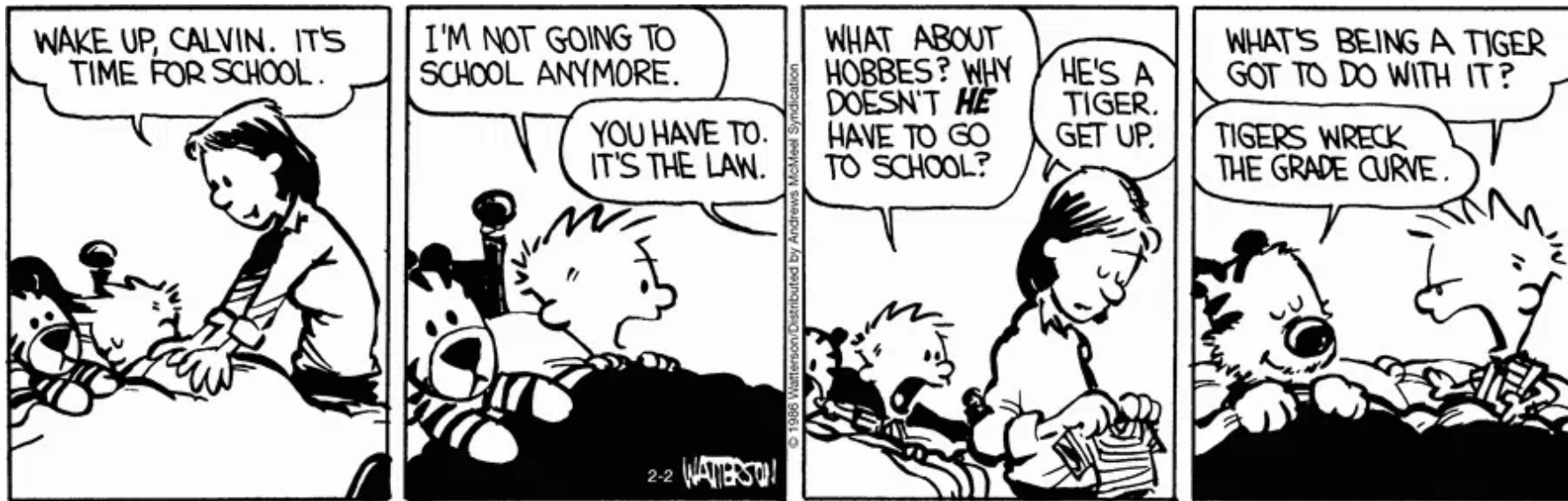


Day 1: Welcome to class!



INFO 1100: Introduction to Programming

Today's Plan

- Introductions
- Learning to program
- Why learn to program?
- Why this class?
- Why Python?
- Class expectations + structure

Introductions

Introductions

Who am I?

Introductions

Who am I?



Rebecca M. M. Hicke
(she / her / hers)

Introductions

Who am I?



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(she / her / hers)

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MS in Computer Science (2024)

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Introductions

Who are you?

Introductions

Who are you?



- Name
- Pronouns
- Year
- Where are you?
- What are you doing this summer?

Learning to program

Learning ~~to program~~

**computational
problem solving**

Learning ~~to program~~

**computational
problem solving**

You are learning *a new logical
framework*

Learning ~~to program~~

**computational
problem solving**



It's hard!

You are learning *a new logical
framework*

Learning ~~to program~~ **computational problem solving**

It's hard!

You are learning *a new logical
framework*

Give yourself
grace!

Why learn to program?

Why learn to program?

*It's an incredibly powerful **tool**!*

Why learn to program?

*It's an incredibly powerful **tool**!*

Robotics

Building assistants
to support in daily
tasks like cooking,
etc.

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Biology

AlphaFold, an AI
program, predicts
protein shapes

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Building assistants to support in daily tasks like cooking, etc.

Library Sciences

Preserving and organizing digital materials for future generations

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AlphaFold, an AI program, predicts protein shapes

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AlphaFold, an AI program, predicts protein shapes

Media Studies

Study how visual and audio pacing has changed in American sitcoms

Why learn to program?

*It's an incredibly powerful **tool**!*

Robotics

Building assistants to support in daily tasks like cooking, etc.

Library Sciences

Preserving and organizing digital materials for future generations

Astronomy

Create simulations of systems that humans can't study up close

Biology

AlphaFold, an AI program, predicts protein shapes

Media Studies

Study how visual and audio pacing has changed in American sitcoms

Why learn to program?

Why learn to program?



in the age of AI

Why learn to program?

^
in the age of AI

Learn how to use
the tool effectively

Why learn to program?

^
in the age of AI

Learn how to use
the tool effectively

Know what is
possible and how to
ask for it

Why learn to program?

in the age of AI

Learn how to use
the tool effectively

Know what is
possible and how to
ask for it

Determine whether
generated code does
what you wanted

Why learn to program?

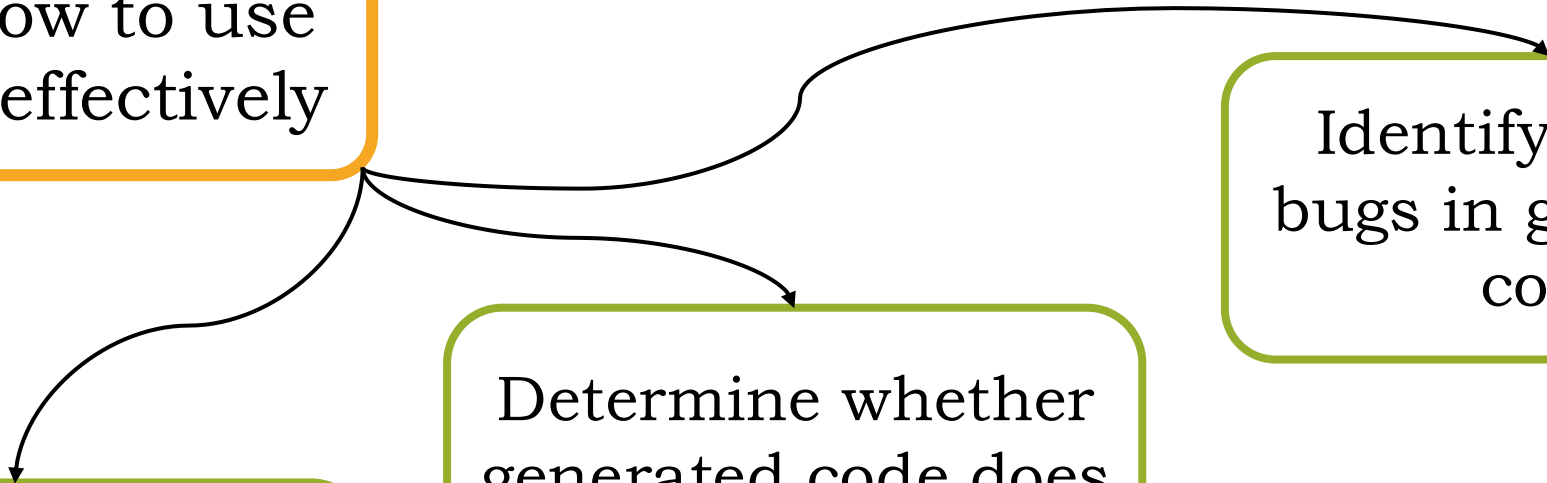
^
in the age of AI

Learn how to use
the tool effectively

Know what is
possible and how to
ask for it

Determine whether
generated code does
what you wanted

Identify and fix
bugs in generated
code



Why *this* class?

Why *this* class?

We'll focus on...

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We'll focus on...

- learning to *think* like a programmer

Why *this* class?

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- *practical* skills for using programming

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We'll focus on...

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- *practical* skills for using programming
- *understanding* and *describing* code

Why *this* class?

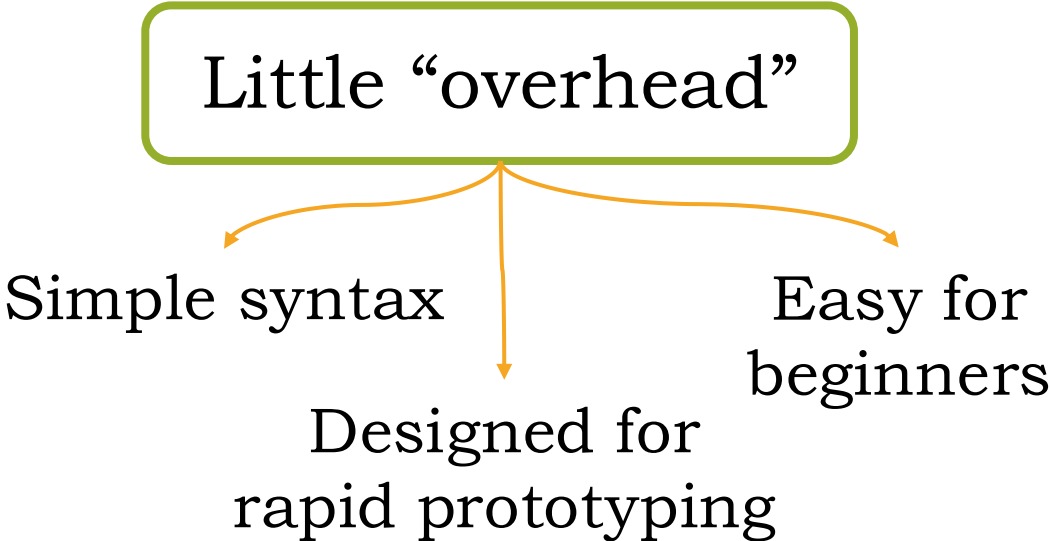
We'll focus on...

- learning to *think* like a programmer
- *practical* skills for using programming
- *understanding* and *describing* code
- how to *use AI tools* for programming

Why **Python**?

Why **Python**?

Little “overhead”



```
graph TD; A[Little "overhead"] --> B[Simple syntax]; A --> C[Designed for rapid prototyping]; A --> D[Easy for beginners];
```

Simple syntax

Designed for
rapid prototyping

Easy for
beginners

Why **Python**?

Little “overhead”

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graph TD; A[Little "overhead"] --> B[Simple syntax]; A --> C[Designed for rapid prototyping]; A --> D[Easy for beginners];
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Simple syntax

Designed for
rapid prototyping

Easy for
beginners

Highly applicable

```
graph TD; A[Highly applicable] --> B[Useful packages:  
Numpy, Pandas,  
SciPy]; A --> C[Used frequently  
for ML];
```

Useful packages:
Numpy, Pandas,
SciPy

Used frequently
for ML

Why **Python**?

Little “overhead”

Simple syntax

Easy for
beginners

Designed for
rapid prototyping

Highly applicable

Useful packages:
Numpy, Pandas,
SciPy

Used frequently
for ML

A modern language

Web applications

Mobile apps

Class expectations + structure

Class expectations + structure

6 weeks

1	2	3	4	5	6
----------	----------	----------	----------	----------	----------

Class expectations + structure

6 weeks

1	2	3	4	5	6
----------	----------	----------	----------	----------	----------

3 credits = 18.75 *hours/week*

Class expectations + structure

6 weeks

1	2	3	4	5	6
----------	----------	----------	----------	----------	----------

3 credits = 18.75 *hours/week*

we're going to move ***fast***

Class expectations + structure

6 weeks

1	2	3	4	5	6
----------	----------	----------	----------	----------	----------

Talk to me!

Class expectations + structure

6 weeks

1	2	3	4	5	6
----------	----------	----------	----------	----------	----------

Talk to me!

weekly meetings

email

office hours

anonymous form

Class expectations + structure

Assignments

Readings

Guided reading questions

Labs

Assignments

Quizzes

Final Project

Class expectations + structure

Assignments

Readings (18)

Guided reading questions (18)

Labs (19)

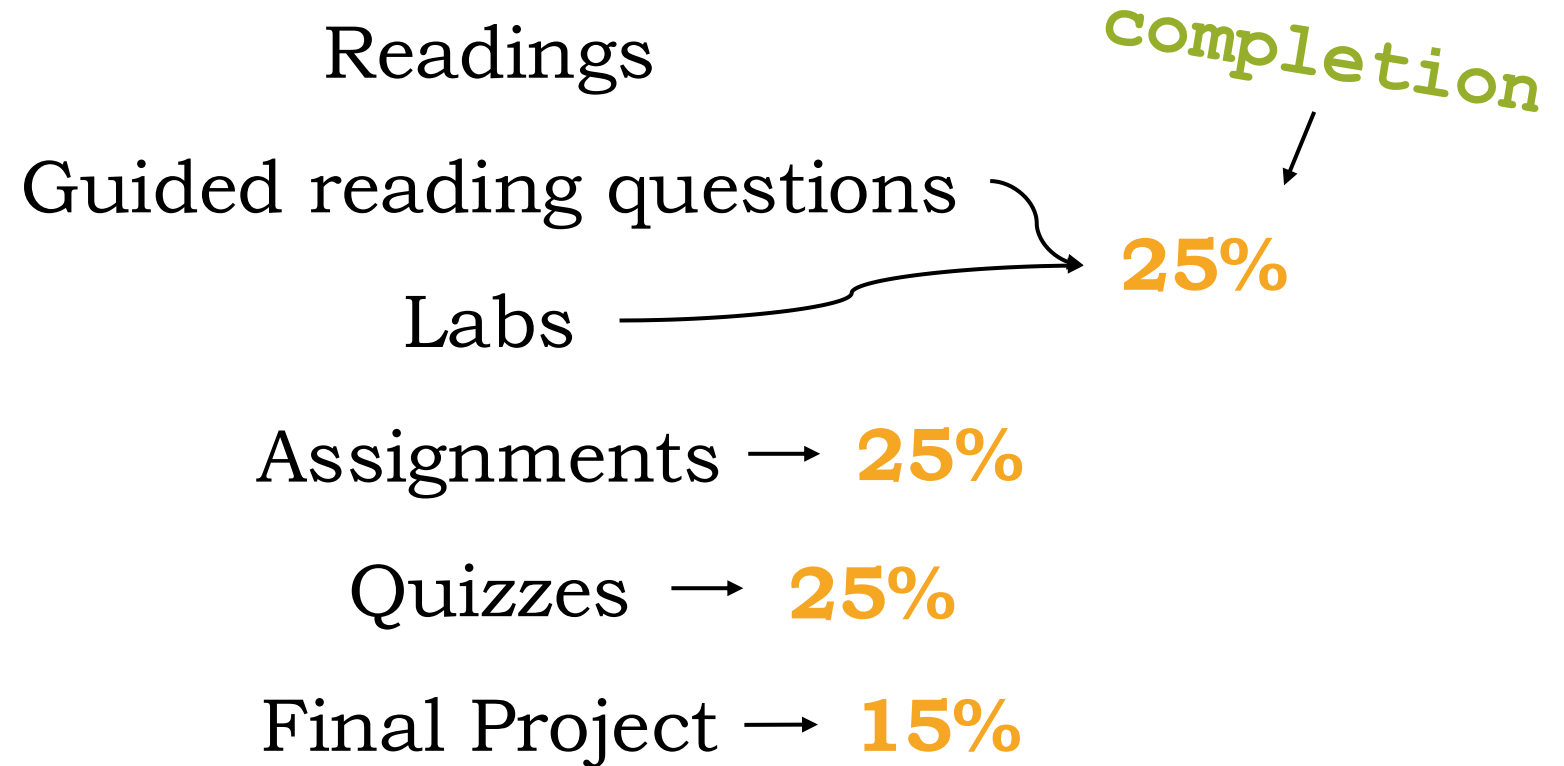
Assignments (4)

Quizzes (5)

Final Project (1)

Class expectations + structure

Assignments



Class expectations + structure

Assignments

Readings

Guided reading questions

Labs

Assignments → 25%

Quizzes → 25%

Final Project → 15%

completion

25%

+ Participation
10%

Class expectations + structure

Assignments

one-on-one
meetings



+ Participation
10%

Readings

Guided reading questions

Labs

completion



25%

Assignments → **25%**

Quizzes → **25%**

Final Project → **15%**

Class expectations + structure

Assignments

$\frac{2}{3}$ grade: completion + code correctness

$\frac{1}{3}$ grade: oral assessment

Class expectations + structure

Quizzes

$4/10$ grade: content question

$4/10$ grade: “explain the code” question

$2/10$ grade: survey on weekly content
(*completion*)

Class expectations + structure

Final Project

$\frac{5}{15}$ grade: project proposal

$\frac{5}{15}$ grade: completion + code correctness

$\frac{5}{15}$ grade: oral assessment

Class expectations + structure

Late Work

3 slip days

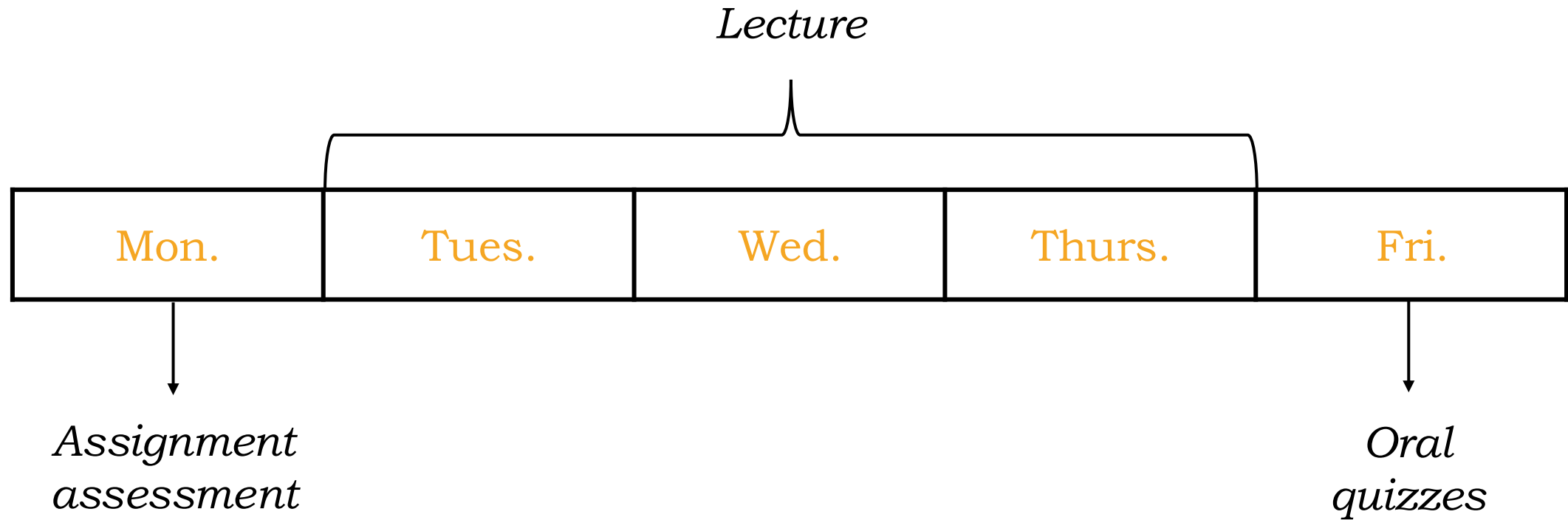
Class expectations + structure

A normal week...

Mon.	Tues.	Wed.	Thurs.	Fri.
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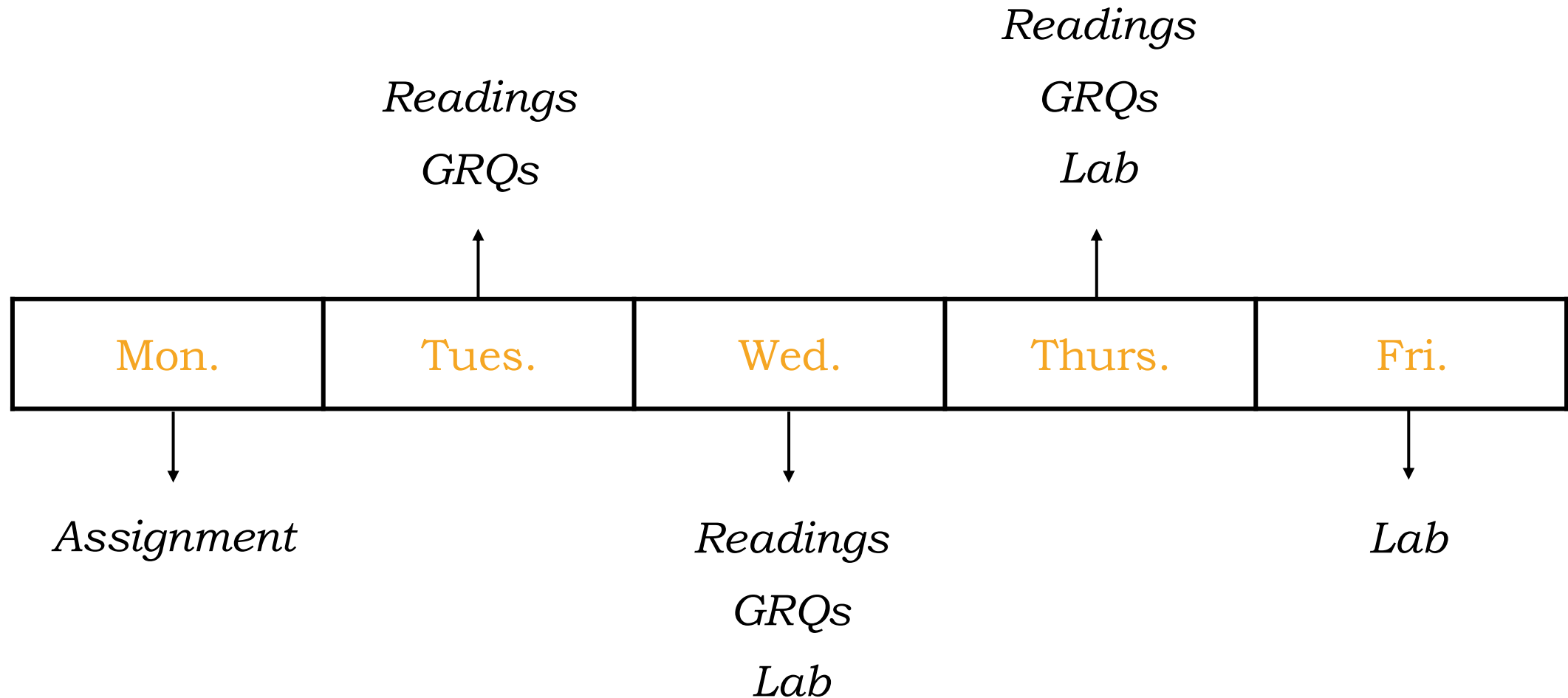
Class expectations + structure

Activities



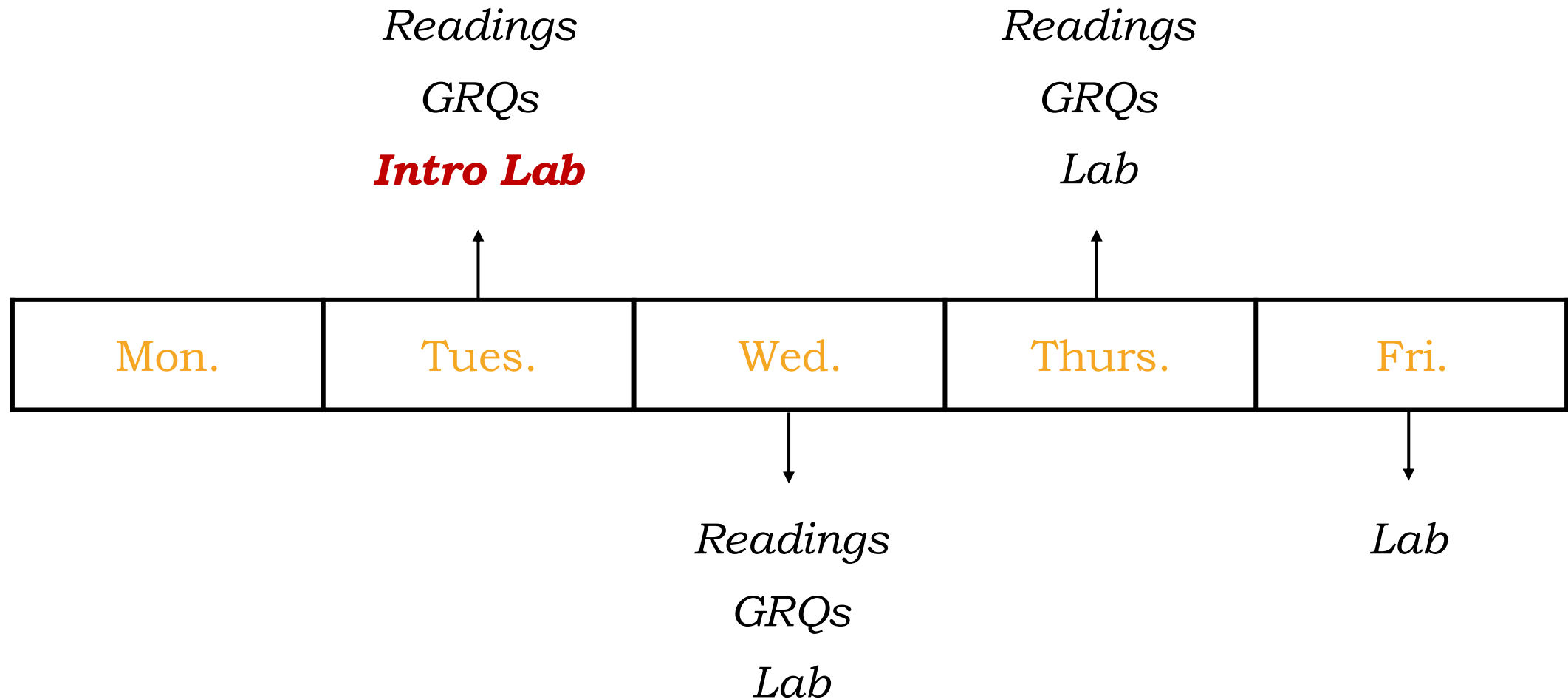
Class expectations + structure

Deadlines



Class expectations + structure

Deadlines



This week

Class expectations + structure

*Why so many
assignments?*

Class expectations + structure

High structure
classes

Class expectations + structure

High structure
classes

Frequent assignments are
frustrating but lead to:

- long-term learning
- less outside studying
- lower failure rates
- higher grades

Class expectations + structure

Online material: Canvas

Lectures: This zoom!

Textbooks: Free – linked on Canvas and syllabus

Office hours:

T (10:30-11:30am, 2:00-3:00pm)

W (10:00-11:00am, 4:00-5:00pm)

Th (9:00-10:00am, 2:00-3:00pm)

Questions?

For tomorrow:

(06/23 @ 11:30am)

- Lab 1
- Readings
Think Python – Chapter 1
- GRQs